PRODUCT DESIGN REQUIREMENTS

Super8 Bedford

Hotel Employee Management Application

Document Version: 1.0

Prepared: 2025

Classification: Internal / Confidential

Product Name HotelShift Pro	**Platform** Mobile Web Application (iOS & Android)
Client Super8 Bedford Hotel	**Document Type** Product Design Requirements (PDR)

1. Executive Summary

HotelShift Pro is a production-grade, mobile-first web application designed specifically for employees of Super8 Bedford Hotel. The application provides a comprehensive, real-time system for clock-in/clock-out management, automated paystub estimation, meal voucher scanning and tracking, and multi-role shift scheduling.

The system caters to two primary user roles: Employees (front desk staff, shuttle drivers, and general hotel workers) and Administrators (hotel managers). Every employee receives a unique login, and the platform intelligently calculates earnings based on hours worked, tips, room bonuses, meal voucher commissions, and hotel occupancy incentives.

2. Project Overview

2.1 Product Scope

Application Name	HotelShift Pro
Property	Super8 Bedford Hotel
Target Users	Hotel Employees (Front Desk, Shuttle Drivers), Hotel Managers/Admins
Platform	Mobile Web Application (Progressive Web App)
Base Hourly Rate	$10.00 per hour (all employees)
Architecture	Multi-user, role-based access control (Employee & Admin)
Primary Goal	Automate time tracking, paystub estimation, bonus calculation, and meal voucher management

2.2 Business Context

Hotel employees at Super8 Bedford currently manage time tracking and earning calculations manually, which introduces errors, delays, and employee dissatisfaction. HotelShift Pro digitizes and automates the entire compensation lifecycle, from clocking in/out to viewing estimated paystubs, handling tip recording, and scanning meal vouchers issued to distressed airline passengers.

2.3 Key Stakeholders

Stakeholder	Role	System Access
Hotel Manager	Admin - full system control	Admin Panel
Front Desk Employee	Clock in/out, view pay, scan vouchers	Employee App

Stakeholder	Role	System Access
Shuttle Driver Employee	Multi-session clock in/out, view shifts	Employee App
Hotel Guests	Indirect - meal voucher recipients	None (QR code only)

3. User Roles & Authentication

3.1 Employee Role

- Unique username and password per employee
- Can view only their own time records, earnings, and vouchers
- Can clock in and clock out for their assigned shift
- Can scan and log meal vouchers via QR code
- Can share meal voucher access with specific team members
- Can view full shift history and estimated paystubs

3.2 Admin / Manager Role

- Full visibility of all employee records and schedules
- Ability to edit, override, or approve clock-in/out entries
- Manage employee profiles, roles, and pay rates
- View all meal voucher logs and transaction history
- Generate and export payroll reports
- Manage hotel sold-out status (triggers $25 bonus)
- View real-time dashboard of who is clocked in

3.3 Authentication Requirements

Requirement	Details
Unique Login	Each employee has a unique username + password
Session Management	JWT tokens with secure cookie storage, auto-expiry after inactivity
Password Policy	Minimum 8 characters, must include number and special character
Role-Based Routing	Employees redirect to Employee Dashboard, Admins to Admin Panel
Forgot Password	Email-based reset flow
Multi-Device	Responsive mobile web app; works on any mobile browser

4. Shift Structure & Scheduling

4.1 Standard Shifts

The application supports four defined shift types, with the ability for employees to clock in early or late while still correctly attributing hours to the correct shift.

Shift Name	Start Time	End Time	Duration	Notes
Morning	7:00 AM	4:00 PM	9 hours	Standard front desk shift
Evening	4:00 PM	12:00 AM	8 hours	Overlaps with night
Night	12:00 AM	7:00 AM	7 hours	Overnight shift
Shuttle	9:00 PM	9:00 AM	Flexible	Multiple clock-in/out sessions allowed

4.2 Shift Logic Rules

- If an employee clocks in before their shift starts, extra time is tracked and labeled as overtime/early
- Shuttle drivers may have multiple clock-in and clock-out sessions within a single calendar day
- Each clock-in/clock-out pair is displayed individually on the dashboard
- Total daily hours are computed as the sum of all session durations for that day
- An employee may work multiple roles across different days (e.g., front desk on Monday, shuttle on Tuesday)
- Role for each shift is selected at clock-in time

4.3 Multi-Role Support

A single employee may be assigned to different roles on different days. The system tracks:

- Role type: Front Desk or Shuttle Driver
- Applicable bonus structures per role
- Separate shift display for each day and role

5. Clock-In / Clock-Out System

5.1 Clocking In

Step	Action	Details
1	Employee logs in to app	Redirected to Employee Dashboard
2	Tap 'Clock In' button	Timestamp recorded with current date/time
3	Select shift role	Front Desk or Shuttle Driver
4	Select shift type	Morning, Evening, Night, or Shuttle
5	Confirmation shown	Clock-in time displayed on dashboard

5.2 Clocking Out

Step	Action	Details
1	Tap 'Clock Out' button	Timestamp recorded
2	Optional: Enter tips received	Dollar amount input field
3	Optional: Enter room bonuses	Number of rooms sold above base price
4	Session saved	Hours, earnings, and bonuses calculated
5	Dashboard updates	Running total updated instantly

5.3 Shuttle Driver Multi-Session Logic

Shuttle drivers operate on flexible schedules, making multiple trips per shift window. The system allows:

- Unlimited clock-in/clock-out pairs per day for shuttle role
- Each trip session is shown as a separate row on the dashboard with its own start/end time
- Total hours for the day aggregate across all sessions
- Paystub estimation accounts for all sessions in a pay period

5.4 Dashboard Time Display

The main employee dashboard displays:

- Current clock-in status and live timer
- All clock-in/clock-out pairs for the current day, each in its own card

- Total hours worked today
- Cumulative weekly hours and estimated earnings
- Current bonus/tip totals

6. Compensation & Earnings Calculation

6.1 Base Pay

Hourly Rate	$10.00 per hour (all employees, all roles)
Pay Frequency	Weekly (configurable by admin)
Overtime Rule	Admin-configurable; standard hours tracked
Pay Calculation	Total hours x $10.00 = Base Pay

6.2 Bonus & Incentive Structure

Bonus Type	Trigger	Amount	Who Qualifies
Room Upsell Bonus	Room sold for $5 more than base rate	$1 per $5 upsold	Front Desk Employee
Meal Voucher Bonus	Each breakfast voucher scanned/sold	$3 per voucher	Front Desk Employee
Hotel Sold-Out Bonus	Hotel reaches 100% occupancy	$25 flat bonus	All clocked-in employees
Tip Income	Cash tip received from guest	100% of tip amount	Any employee (self-reported)

6.3 Room Upsell Calculation Example

Room base price: $100 (before tax).
Room base price: $100. The guest pays $140. Upsell amount: $40. Bonus: $40 / $5 = $8.00.
The formula applied is: Bonus = floor((Guest Price - Base Price) / 5) x $1

6.4 Estimated Paystub Calculation

The paystub is an estimate updated in real-time and is NOT a final payroll document. It includes:

- Gross Pay = (Total Hours x $10.00)
- Room Bonus = Sum of all upsell bonuses for the period
- Meal Voucher Bonus = (Count of vouchers scanned x $3) - sometimes onces the QR scanned then the employee enter into system it get fails because it was already used in Airport, so that time the count must not get scan
- Sold-Out Bonus = Number of sold-out events x $25
- Tips = Total self-reported tips
- Total Estimated Earnings = Gross Pay + All Bonuses + Tips

6.5 Paystub Display

Employees can view their estimated paystub in a dedicated tab showing:

- Pay period date range
- Hourly breakdown by day and shift
- Itemized bonuses and tips
- Total estimated gross earnings
- Printable/shareable paystub summary

7. Meal Voucher System

7.1 Overview

Airport-distressed passengers are provided meal vouchers by their airline or a third-party provider. These vouchers arrive as QR codes and contain guest/card information. The employee scans the QR code at the front desk to process the voucher.

7.2 Meal Voucher QR Scanner

Feature	Description
QR Code Scanning	Native mobile camera-based QR code scanner built into the app
Data Extraction	Parses voucher data: guest name, card/payment details, voucher amount, checkin code
Room Association	Employee enters room number the voucher is associated with
Passenger Count	Employee enters number of passengers covered by the voucher
Confirmation	Summary screen shown before submission; employee confirms and submits
Auto-Bonus Credit	System automatically credits $3 bonus per voucher to the scanning employee

7.3 Voucher Data Fields Captured - this must be the smoothest functionality in the application.

- Voucher ID / QR code reference
- Onces the QR is scanned then the text we get is ‘M1MC5567665957881786/1227’ so in which M1MC says it’s a mastercard, then 5567.. says card number and then after slash 1227 is the expired date of card.
- Card/payment details (parsed from QR, encrypted at rest)
- Voucher value / type (breakfast only)
- Room number (entered by employee)
- Number of passengers (entered by employee)
- Scan date and time (auto-stamped)
- Employee who scanned the voucher

7.4 Voucher Management Tab

All scanned vouchers are stored and accessible in a dedicated 'Vouchers' tab with:

- List view of all vouchers scanned by the logged-in employee

- Search and filter by date, room number, or passenger count
- Detail view for each voucher showing all captured fields
- Admin view: all vouchers across all employees

7.5 Voucher Sharing / Access Control

Employees can selectively share their meal voucher records with specific team members:

- Access control list: employee selects specific colleagues from a list
- Granular toggle: choose which voucher records to share
- Shared employees can view (read-only) the shared records in their Vouchers tab
- Access can be revoked at any time
- Admin always retains full access to all voucher data

8. Screen & Feature Specifications

8.1 Employee App Screens

Screen	Purpose	Key Components
Login	Authentication	Username, password, forgot password link
Dashboard	Home screen	Clock in/out button, live timer, today's sessions, quick stats
Shift History	Past shifts	Calendar view + list of all clock-in/out pairs with hours
Paystub	Estimated earnings	Pay period selector, itemized earnings breakdown, totals
Meal Vouchers	Voucher management	QR scanner, voucher list, detail view, sharing settings
Profile	Account settings	Name, contact info, role, password change
Notifications	Alerts	Sold-out bonus alerts, pay period summaries, admin messages

8.2 Admin Panel Screens

Screen	Purpose	Key Components
Admin Dashboard	Overview	Live clocked-in employees, today's summary stats
Employee Management	Manage staff	Add/edit/deactivate employees, assign roles
Time Records	All time logs	Filterable table of all clock-in/out events; edit/override
Payroll Report	Pay reporting	Export by pay period, per employee or all staff
Voucher Log	All vouchers	Full voucher history across all employees
Sold-Out Control	Occupancy	Toggle hotel sold-out status; triggers bonus for all active staff
Settings	System config	Pay periods, shift definitions, hourly rate, bonus toggles

8.3 Mobile UX Requirements

- Fully responsive design optimized for screens 375px - 428px wide
- Touch-friendly buttons (minimum 44x44px tap targets)
- Offline-capable clock-in (queues sync when connection restored)
- Dark/light mode support
- Native-feel animations and transitions
- Bottom navigation bar for primary sections (Dashboard, Shifts, Paystub, Vouchers, Settings)

9. Technical Architecture

9.1 Recommended Tech Stack

Layer	Technology	Rationale
Frontend	React Native Web / Next.js (PWA)	Single codebase, mobile-first PWA, installable on home screen
State Management	Zustand or Redux Toolkit	Predictable state for real-time clock and earnings updates
Backend API	Node.js + Express or Fastify	Lightweight REST API with WebSocket for live updates
Database	PostgreSQL	Relational data for shifts, employees, earnings, vouchers
Authentication	JWT + bcrypt	Secure token auth with hashed passwords
QR Scanning	react-qr-reader / ZXing	Mobile browser-based QR code parsing
Hosting	AWS	Scalable, zero-downtime deployment
Encryption	AES-256 for card data	Payment details from voucher QR encrypted at rest

9.2 Data Models (High-Level)

Employee

- id, name, email, role (employee | admin), hourly_rate, created_at

Shift Session

- id, employee_id, clock_in, clock_out, role_type, shift_type, tips, notes

Bonus Event

- id, employee_id, type (room | voucher | soldout | tip), amount, timestamp, reference_id

Meal Voucher

- id, scanned_by, scan_time, room_number, passenger_count, voucher_id, guest_name, encrypted_card_data, raw_qr_data

Voucher Access

- id, voucher_id, owner_employee_id, shared_with_employee_id, can_view

9.3 Security Requirements

- All API endpoints protected by JWT authentication
- Role-based middleware: admin routes inaccessible to employee accounts
- Payment/card data from QR codes encrypted with AES-256 before database storage
- HTTPS enforced on all environments
- QR scan data never logged in plain text
- Session timeout after 30 minutes of inactivity

10. Non-Functional Requirements

Requirement	Target
Performance	Dashboard loads within 1.5 seconds on 4G mobile connection
Uptime	99.5% availability; critical clock-in/out works offline with sync
Scalability	Supports up to 50 concurrent employee users
Data Accuracy	Paystub calculations accurate to the cent
Browser Support	iOS Safari 15+, Android Chrome 90+
Accessibility	WCAG 2.1 AA compliance; font size minimum 14px
Backup	Daily automated database backups; 30-day retention
Audit Log	All clock-in/out events and admin overrides are immutably logged

11. Acceptance Criteria

11.1 Core Feature Acceptance

Feature	Acceptance Criteria
Clock In/Out	Employee can clock in and out; timestamp saved within 1 second; session visible on dashboard immediately
Multi-session Shuttle	Shuttle driver can clock in/out 5+ times in one day; all sessions shown with correct times and totals
Paystub Calculation	Estimated paystub correctly calculates base pay + all bonus types; matches manual calculation to the cent. Always show each and every distribution with details
Room Bonus	Entering a room sold for $100 base $110 yields $2 bonus; $140 yields $8 bonus(i.e. Every above 5$ from base price, employee gets 1$)
Voucher Scanner	QR code scanned; data auto-populated; room number and passenger count added; $3 bonus credited But sometimes the Meal voucher get declines after 1 hr so that time user should have option to edit the number of meal voucher, because it's linked with paycheck estimate.
Sold-Out Bonus	Admin triggers sold-out; all currently clocked-in employees receive $25 bonus entry
Voucher Sharing	Employee shares voucher with colleague; colleague sees it in read-only mode; revoke removes access
Admin Override	Admin can edit any clock-in/out time; change is logged with admin's ID and timestamp
Role Separation	Admin cannot access employee view; employee cannot access admin panel

12. Out of Scope (v1.0)

- Direct payroll processing or payment disbursement
- Integration with hotel PMS (Property Management System) in v1.0
- Guest-facing features or portal
- Native iOS / Android app builds (v1.0 is PWA only)
- Multi-property / multi-hotel support
- HR document management (W-2, tax forms)

13. Risks & Mitigations

Risk	Likelihood	Impact	Mitigation
QR code format changes by airline	Medium	High	Build flexible QR parser; admin can manually enter voucher data
Offline clock-in sync conflicts	Low	Medium	Last-write-wins with audit log; admin review queue for conflicts
Employee gaming the sold-out bonus	Low	Medium	Admin-only sold-out trigger; immutable audit log
Card data breach from voucher QR	Low	High	AES-256 encryption, never display raw card data in UI
Wrong shift attributed to early clock-in	Medium	Medium	Employee selects shift type at clock-in; admin can override

14. Glossary

Term	Definition
PDR	Product Design Requirements - this document
Clock-In	The act of recording the start of a work session
Clock-Out	The act of recording the end of a work session
Meal Voucher	A QR-coded voucher issued to distressed airline passengers for hotel breakfast
Sold-Out Bonus	A $25 bonus triggered when the hotel reaches 100% occupancy
Room Upsell	When a guest pays more than the base room rate; generates $1 per $5 overpayment bonus
Shuttle Session	A single clock-in / clock-out pair for a shuttle driver trip
PWA	Progressive Web App - a web app installable on mobile home screens
Admin	A manager-level user with full system access
JWT	JSON Web Token - used for secure user authentication
AES-256	Advanced Encryption Standard 256-bit - used to encrypt sensitive voucher card data

15. Document Approval & Sign-Off

This PDR must be reviewed and approved by all relevant stakeholders before development commences.

Role	Name	Signature	Date
Hotel Manager / Product Owner			
Lead Developer			
QA Lead			
Operations Lead			

Document prepared with Anthropic AI assistance. All business logic and requirements based on stakeholder input.